

Project Report: Identification of canonical neural events during continuous gameplay of an 8-bit style video game

Sana Hafeez, Hadi Ismail, Okan Mazlum

SS 2026

Introduction

This report documents an attempt to reproduce the main findings of *Identification of canonical neural events during continuous gameplay of an 8-bit style video game* by Cavanagh and Castellanos (2016), for the module *Signal processing and Analysis of human brain potentials (EEG)*. [1] We use the study's publicly shared EEG dataset, OpenNeuro ds003517. [2] The accompanying project repository provides the code and reproduction instructions. [3]

The main question of this project is to see whether we can reproduce the results of the paper using a different preprocessing and analysis pipeline, which would strengthen the robustness of the findings in the original paper.

Original paper

The motivation of the original paper is the issue of generalizability in classical EEG experiments. In order to minimize artifacts, EEG tasks are often designed with isolated stimuli in carefully controlled environments. These however can be difficult to interpret in a complex, continuous environment. The authors developed the space-shooter game *Escape from Asteroid Axon* to study these complex events while retaining precise event timing. Before playing, subjects completed two established exemplar tasks: oddball target detection and gambling wins/losses. The authors compared the EEG responses statistically and trained subject-specific classifiers on these exemplar tasks, then transferred their learned patterns to gameplay. [1]

The main finding was that wall crashes, enemy crashes and missile hits showed a transfer bias toward oddball-rare patterns during the P3 period. Missile hits also resembled gambling wins, whereas crashes did not reliably resemble gambling losses. The authors interpreted these results as evidence that some laboratory signatures associated with salience and reward generalize to gameplay; the expected punishment-related result (gambling loss transfers to crashes) could not be established. [1]

Dataset description

The dataset contains 17 subjects (11 men; ages 18–39, mean 20.94 years), reported as right handed. Each subject has two runs under the BIDS task name ContinuousVideoGamePlay. Run 01 contains oddball and gambling tasks; run 02 contains gameplay. Full recording durations average 777.78 s and 1,976.49 s respectively and include periods outside the active tasks. EEG was sampled at 500 Hz with CPz as acquisition reference. There are 63 recorded EEG channels plus bipolar VEOG and HEOG; CPz is not independently recorded. All of the supplied electrode positions match the BESA standard-10-5-cap385.elp montage [2]

Oddball

Participants mentally counted rare enemy-spacecraft images without a button response. The intended design contained 160 standard and 40 rare images. Each image was shown for 500 ms, followed by a random 1–2 s interval. Rare events occurred after at least three but before ten standards. [1] The duration of this task averaged 6.68 min across the 17 subjects.

Gambling

Participants chose one of two doors using the left/right gamepad controls. Each trial resulted in a win of 75 credits or a loss of 60 credits. The intended design contained 40 wins and 40 losses. The doors remained on screen for up to 4 s or until a choice. Feedback was then presented immediately for 500 ms, followed by a 1 s interval. The paper gives a mean task duration of 2.86 min. [1]

Escape from Asteroid Axon

Participants steered a ship and fired missiles, avoided wall/enemy crashes, collected stars to restore health and collected ammunition. An empty health bar ended the round; another round could be started by a button press. Game events include shots, star collections, ammunition collection, wall crashes, enemy crashes and missile hits. Gameplay was intended to last 30 min. [1]

Dataset exceptions and recording boundaries

- All 17 subjects have 200 oddball events, but some early recordings contain 37–44 rare events rather than exactly 40.
- Subjects 002, 003 and 005 have no gambling-outcome events, which is also reported in the original paper.
- Subject 001 has only 18 win and 17 loss events; subject 006 has 120 wins and 120 losses. The remaining subjects with gambling events have the intended 40 wins and 40 losses.
- Subject 001 has the extended gameplay session: approximately 45.9 min through the last logged game event, increasing the mean duration across subjects to 31 min.
- While all other subjects have a median of 90s of inactivity before actual gameplay starts, for subjects 003 and 008, the first 41 s and 35 s of gameplay are missing respectively. For subject 008, 11 s of their axon gameplay is instead located at the end of their oddball and gambling run

Preprocessing

We use MNE-BIDS-Pipeline because it is the workflow used in the course. The original authors used the MATLAB toolbox EEGLAB. Differences between these software ecosystems affect many implementation details and defaults (e.g. specific ICA implementation or channel interpolation algorithm). This chapter focuses instead on the main consciously made methodological choices and their reasons.

Initial loading attempt

Channel metadata

With a minimal configuration file, an attempt to load the original dataset fails during the pipeline's initial data-quality report. The error is `picks (None, treated as "data") yielded no channels`, consider passing picks explicitly. The reason is that all entries in the original `channels.tsv` type and units columns are `n/a`, making MNE believe there are no data channels for its calculations.

This motivated a preparation script `scripts/prepare_bids.py`, which creates a new “prepared” dataset with corrected metadata. The 63 scalp channels receive type EEG, VEOG and HEOG receive type EOG, and all channels receive units `uV`. The corresponding `*_eeg.json` channel type counts are adjusted accordingly.

Electrode coordinates

With channel types corrected, another warning appears on reexecution of the pipeline: `Other is not an MNE-Python coordinate frame for EEG data and so will be set to 'unknown'`. The original `*_coordsystem.json` declares `EEGCoordinateSystem = "Other"`, which is technically allowed in BIDS, but not correctly defined here. [4]

The supplied MATLAB scripts, `Convert2BIDS_ExAAx.m` and `ExAAx_Preprocess.m`, identify the used montage as `standard_BESA/standard-10-5-cap385.elc`. This allows us to extend the preparation script to additionally correct `*_electrodes.tsv` and `*_coordsystem.json`, transforming them into the corrected MNE head coordinates.

First inspection and visual delay

The prepared dataset allows an initial pipeline execution and inspection of minimally filtered EEG. However, this reveals yet another issue:

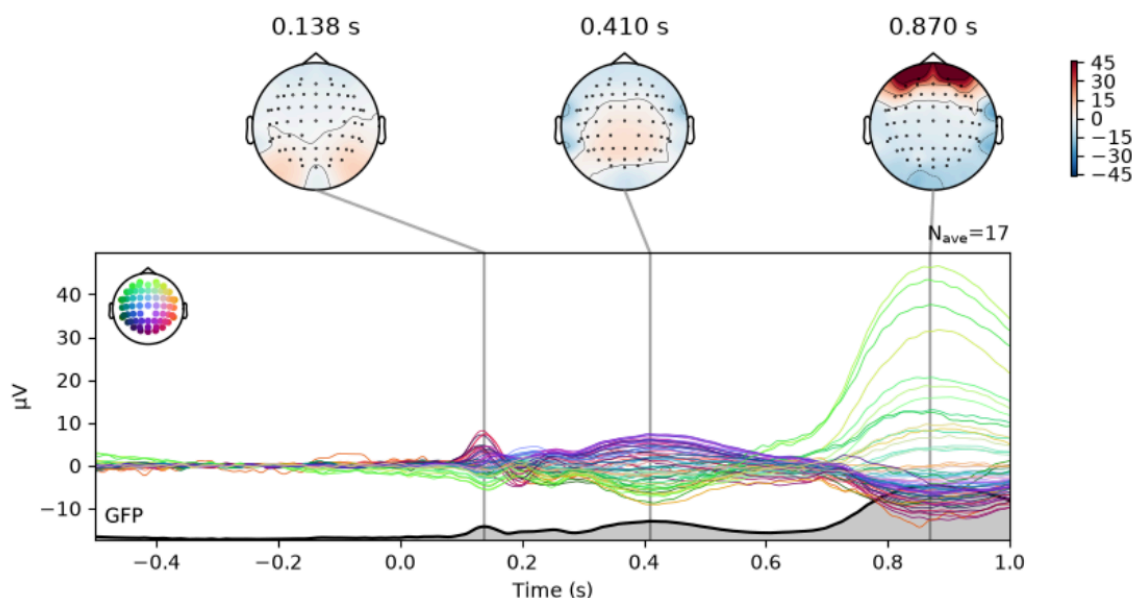


Figure 1: Initial oddball butterfly plot for rare events in subject 001 with minimal filtering

The three observed peaks are the visual P1 (at 0.138s), P3/P300 at 0.410 s and eye artifacts at 0.870 s. However, all of the peaks are slightly too late - the P1 peak would usually be expected around 100ms and P3 at around 300-400ms. The cause of this is the 40ms visual delay also reported by the paper. Since the mne-bids-pipeline does not seem to have a configuration setting to shift event onsets, we also used the preparation script to shift all of the event onsets 40 ms forward, to be in sync with the EEG recording.

Pipeline configuration

The table summarizes the main methodological pipeline differences.

Aspect	Original paper	This project
Bad-channel selection	Four ventral electrodes excluded, then FASTER	PyPREP including RANSAC; all threshold-based flags accepted after inspection; no default exclusions
Frequency filtering	0.01–100 Hz acquisition; 0.1–20 Hz ERP/classification	0.1–100 Hz continuous EEG with a 60 Hz notch; ICA fit at 1–100 Hz; ERP/classification at 0.1–40 Hz
Rereferencing	Initial average reference and CPz reconstruction; subsequent bad-channel interpolation	Average reference over the common non-bad EEG set; no CPz reconstruction

Aspect	Original paper	This project
Combining runs	Supplied MATLAB scripts suggest separate exemplar/gameplay decompositions	One ICA per subject across both runs
ICA training	-2...+2 s epochs around events	Evenly spaced 4 s windows without baseline or channel interpolation before fitting
ICA component selection	VEOG/HEOG correlations with manual verification	ICLabel proposals plus visual review
Bad-epoch handling	FASTER, channel interpolation and epoch rejection	AutoReject after ICA repairs channels and rejects analysis trials

Bad-channel selection

We chose PyPREP because it works with natively MNE data and provides several different channel-quality checks. We use NoisyChannels with default parameters to perform all included checks, including NaN/flat, deviation, high-frequency noise, correlation/dropout, SNR, PSD and RANSAC. [5] In exploratory runs, the flagged channels varied substantially. We therefore fixed the random seed at **2026** and increased RANSAC's `n_samples` to **1,000 random channel subsets** to improve reproducibility and sampling stability.

To check the flagged channels, we built a small inspection tool (`./review-channels` in `01_review_bad_channels/`) using MNE's data browser. It shows the flagged channel alongside neighbouring electrodes using raw or frequency-filtered EEG. For Correlation/dropout and RANSAC, PyPREP provides time intervals that triggered the algorithm to mark a channel as bad, these intervals can also be specifically examined in the tool.

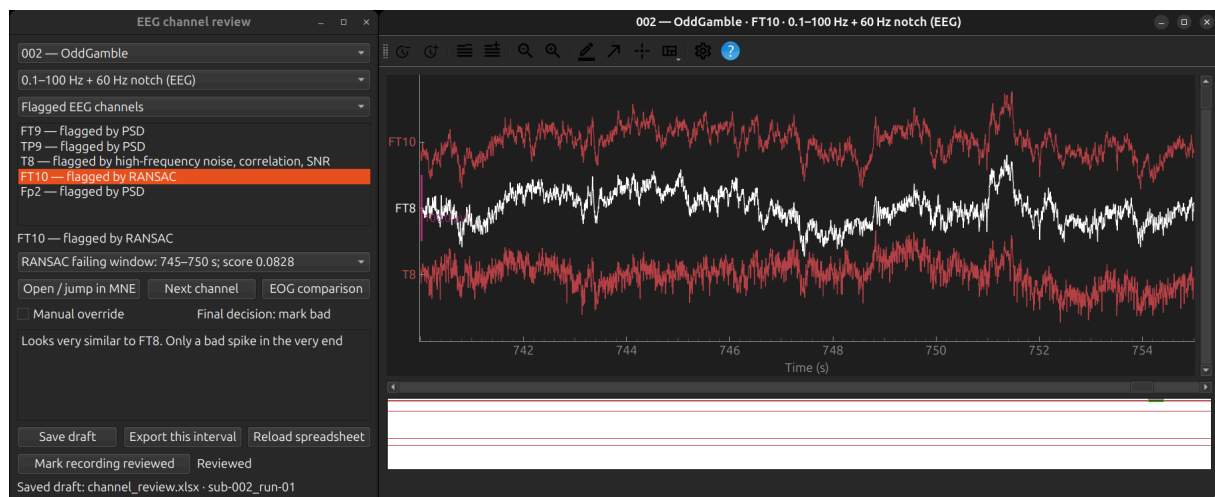


Figure 2: Analysis of run 1 of subject 002 in our tool, highlighting two channels that have been marked as bad

Our visual inspection led to three observations:

- For many flagged channels, the decision was easy to understand: the signal was visibly noisier than that of nearby electrodes (see T8 in Figure 2)
- Some flags were difficult to verify visually, especially RANSAC flags. Its prediction from multiple other electrodes is not necessarily apparent in an isolated trace comparison. (see FT10 in Figure 2, the picture shows the interval that decided on the channel exclusion)

- Some unflagged channels appeared problematic during inspection, but their detector scores lie just below the rejection threshold.

Given the project's scope and our limited experience with manual channel rejection, we adopted all PyPREP threshold-based decisions without manual overrides. Across all recordings, 68 channels were flagged. The median is two flags per recording (range 0–6); six recordings have none. CP2 is flagged most often, in seven recordings. The original paper also reports a median of two interpolated electrodes. [1]

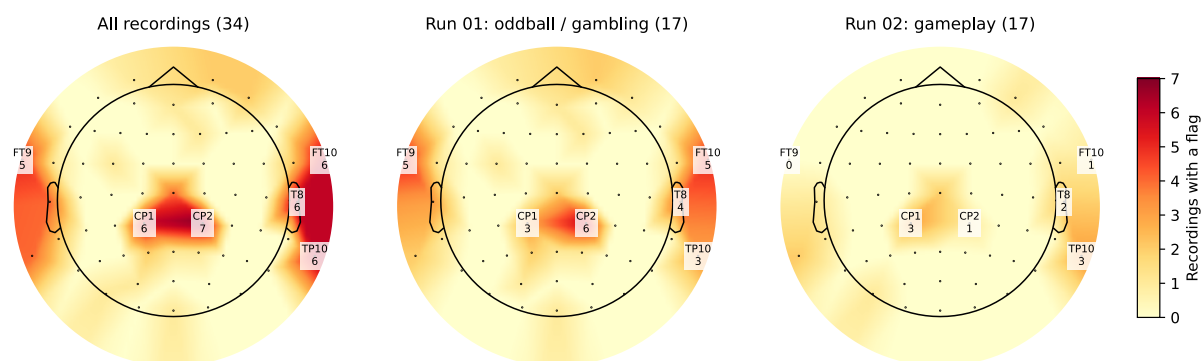


Figure 3: Frequency of PyPREP flags over all 34 recordings and separately by run. All panels use the same count scale. Labels identify electrodes flagged in at least five recordings overall.

The paper excluded FT9, TP9, TP10 and FT10 by default (i.e. 100% exclusion). PyPREP flags 21/136 of these ventral channels (15.4%), which is much lower than the default exclusion, but still high compared with 47/2,006 (2.3%) at the other 59 electrodes.

PyPREP runs in a separate script, `scripts/run_pyprep.py`, after the initial preparation and preliminary frequency-filtering and before the main ICA pipeline execution. The script writes the accepted flags directly into the `*_channels.tsv` sidecars of the prepared dataset. For later steps, it will be required that both runs of each subject share the same set of bad channel lists, which is why we will take the union of both lists here.

Frequency filtering

Continuous EEG is filtered to 0.1–100 Hz with a 60 Hz notch at the original 500 Hz sampling rate. The ERP/classification branch uses a continuously filtered 0.1–40 Hz copy. The 40 Hz low-pass follows the filtering lecture's general recommendation to retain at least 40 Hz; it is the bandwidth of these saved analysis epochs, not only a plotting setting. [6] Wider-band continuous files remain available.

ICA training uses a separate 1–100 Hz copy to follow ICLabel's documented input convention, together with average reference and extended Infomax. [7] This is a recommended input specification rather than a software-enforced prohibition on other bands. The paper specifies 0.01–100 Hz acquisition and 0.1–20 Hz ERP/classification filtering, but no distinct ICA fitting filter. [1] Filtering is performed before epoching.

Rereferencing

The paper describes an initial average reference and CPz reconstruction before ventral-channel removal and FASTER cleaning; the supplied scripts also contain later rereferencing after interpolation. [1] To avoid known bad channels contributing to the average, this project references over the common accepted EEG set after bad-channel selection, excluding EOG and globally bad EEG channels. The same contributing channels and reference are used for ICA fitting and

application. We do not reconstruct CPz. The paper's description does not establish that untreated bad channels remained in its final reference.

Combining runs within subjects

We retain the pipeline's normal behavior of fitting one ICA per subject/task across all included runs. Our configuration explicitly includes runs 01 and 02, equivalent here to selecting all runs. The supplied MATLAB scripts suggest separate exemplar and gameplay processing, but the dataset warns that these scripts may not match the final published analysis, and the paper does not resolve the pooling question.

Joint fitting is supported by the same-session acquisition and common channel layout. Subject 008's gameplay tail illustrates why run labels alone do not perfectly separate the tasks. A common channel/reference policy and ICA transform also make source-task and gameplay data more comparable for transfer analysis. Pooling provides more training EEG when one task loses many windows; it does not guarantee better separation, and assumes sufficiently stable spatial mixing across the session. It cannot restore EEG missing between recordings.

This choice requires the union of each subject's run-specific bad channels and produces one saved ICA model per subject. The installed pipeline fails if the runs' bad-channel lists differ. Each run is filtered and windowed separately before the fitting epochs are concatenated, so windows never cross recording boundaries. For the run-01 reference this additionally excludes PO7 in 009 and CP1 in 012, which were flagged in run 02.

ICA training

An initial pilot used $-2...+2$ s epochs around selected events, following the paper's epoching description. Frequent gameplay events produced extensive overlap, repeatedly presenting the same samples to ICA and increasing processing time. Local AutoReject also removed uneven amounts of exemplar data. We explored fixed windows with a 500 μ V peak-to-peak screen, then returned to fresh local AutoReject for the selected native workflow. These tests changed more than one setting, so their differences cannot be assigned to a single cause.

The final configuration uses `task_is_rest = True` to create evenly spaced, non-overlapping four-second fitting windows and downsamples to 250 Hz with antialiasing. This option selects a time grid; the experiment is not resting-state EEG. The paper also used 250 Hz for classification, but does not clearly specify ICA downsampling. The final approach was much faster and retained recognizable brain and artifact components, although some pilot comparisons showed weaker alpha structure or imperfect component correspondence. It therefore does not establish unchanged decomposition quality in every subject.

The grid is independent of task events: it can include pauses or task transitions within a run. We chose this simpler native behavior instead of custom inference of task blocks, gaps and padding; it does not guarantee balanced task coverage. The detailed comparisons remain in "Everything else". The supplied scripts contain a commented whole-epoch mean-removal step before ICA. Our fitting windows receive no baseline correction and no channel interpolation; the $-200...0$ ms ERP baseline is applied later.

A controlled subject-001 comparison used identical retained data for Picard extended Infomax and regular extended Infomax. Picard took 16 min 47 s, compared with 8 min 49 s for regular Infomax, with closely corresponding components and identical matched automatic exclusions. We therefore retained regular extended Infomax. This single-subject timing comparison is not a general solver benchmark.

Across all 17 subjects, **9,714/11,655 fitting windows (83.35%)** were retained, providing 24.3–54.5 min of training EEG per subject. There are 1,003 components, 56–62 per subject according to available EEG rank after average reference. Post-hoc comparison with event-inferred task spans gives 85.7% oddball, 87.2% gambling and 87.3% gameplay retention. These approximate spans did not control the native grid and do not cover every fitting window.

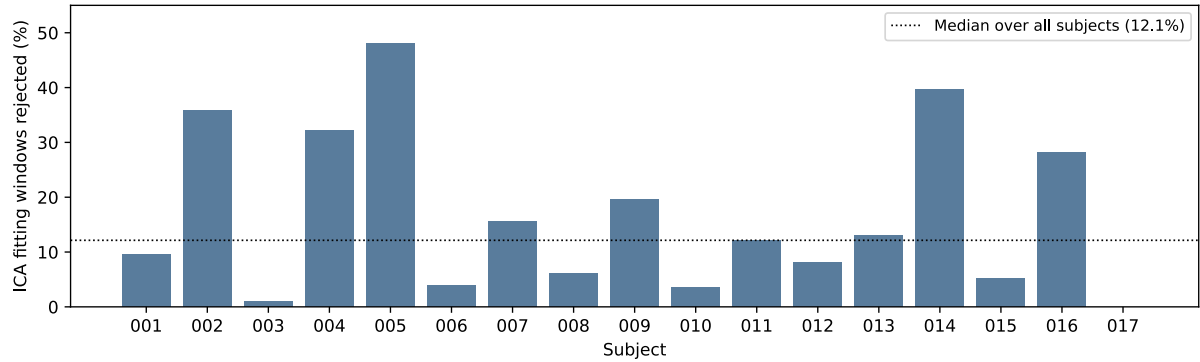


Figure 4: AutoReject rejection of ICA fitting windows by subject. The dotted line is the median across the 17 subject-specific percentages (12.1%); the separately calculated pooled rejection rate is 16.65%.

Rejection varied strongly: subjects 005 and 014 lost 48.2% and 39.8% of fitting windows respectively. Subject 014 retained only 4/33 windows attributed to gambling, but still had more than 24 min of training data overall. Replaying AutoReject reproduced their masks exactly. A counterfactual 500 μ V cutoff would retain 94.7% and 96.7%, including 97/99 oddball windows in 005 and all 33 gambling windows in 014. This check evaluated retention only: no alternative ICA fit was performed for those subjects, so it cannot establish better or worse decomposition. Visual review found their existing decompositions acceptable, and we retained them. Fitting-window losses are distinct from later ERP trial losses.

ICA component selection

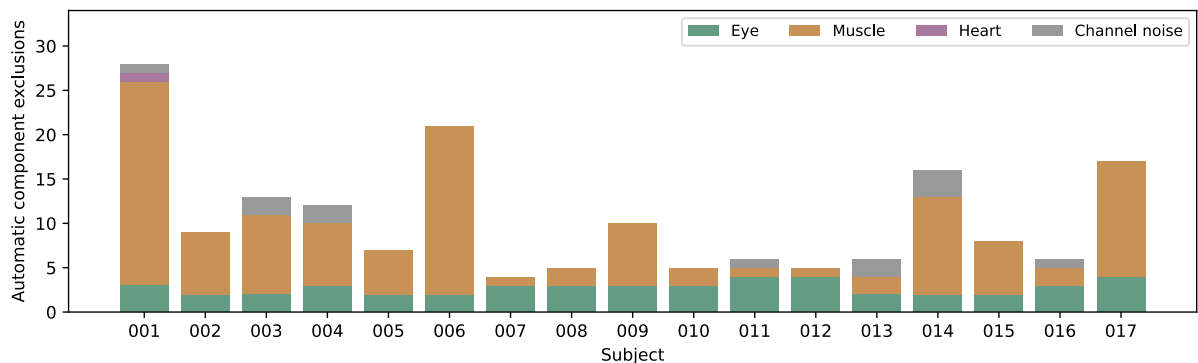


Figure 5: Automatic ICLabel exclusion proposals by subject and category, before manual additions.

ICLabel proposed **178 exclusions**: 118 muscle, 47 eye, 12 channel-noise and one heart component. Among the first ten components (ICA000–ICA009), there are 37 automatic eye proposals across the cohort: **2.18 per subject**. Across all component indices, the mean is $47/17 = 2.76$. The paper reports one blink component per subject, plus two horizontal-eye components in 13 subjects and one in four: also 2.76 ocular components per subject overall, predominantly among its earliest components. [1] ICLabel’s combined eye category does not distinguish blink from horizontal-eye sources, so these are descriptive counts, not a one-to-one validation.

Visual review found recognizable ocular, muscle and brain components, particularly among early components, but also 15 additional artifacts that did not pass the automatic exclusion threshold. We inspected maps, spectra and time courses and manually excluded these additional sources. The following panel illustrates one such decision.

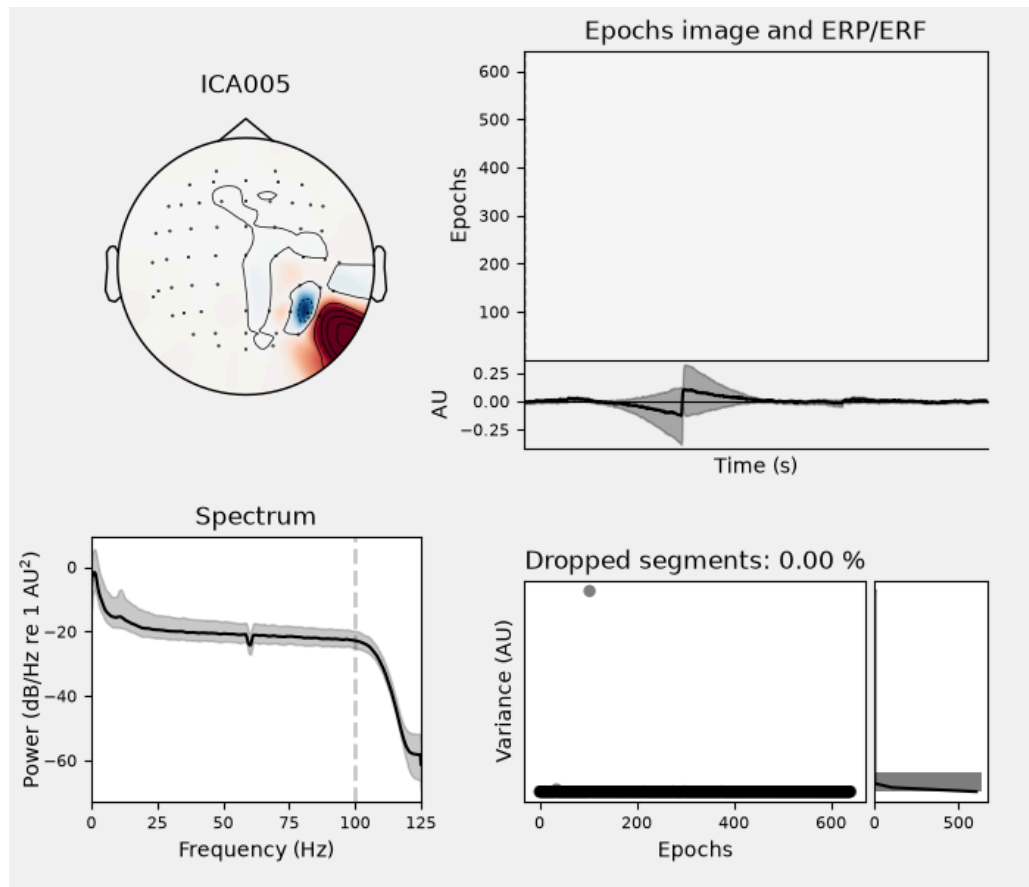


Figure 6: Manual exclusion example: subject 008, ICA005. A focal posterior map and disproportionately large early single-window contribution accompany a 1,641 μV P6 transient. Its channel-noise probability of 0.622 was below the automatic 0.8 threshold; visual review supported exclusion. The fixed-window component average is not a task ERP.

With the manual additions, **193 components** were approved for removal, with a **median of eight per subject**.

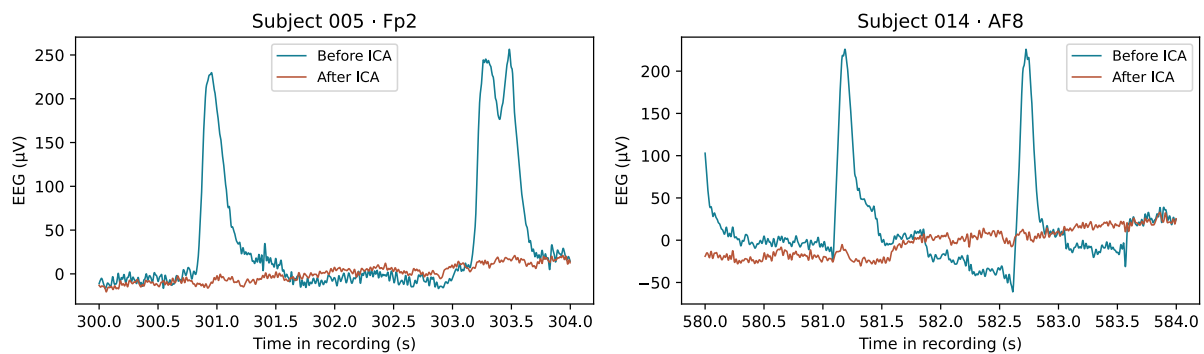


Figure 7: Effect of applying the reviewed ICA models on subjects 005 and 014. Peak-to-peak amplitude in blink-like windows decreases from approximately 273 to 42 μV and 287 to 63 μV respectively. Both curves use 0.1–40 Hz EEG and the same average reference, with individual medians removed for display

The first pipeline pass ends here with saved ICA models, component decisions and reports. A second pass applies those models to EEG prepared for event-related analysis.

Note about EOG channels

[merge this section with Low-variation VEOG channels and significantly shorten it. all the reader needs to know is:

1. EOG is excluded from PyPREP, the EEG ICA fit and the average reference. basically it's not really used for anything in our pipeline
2. the original study does use eog for ica classification
3. we could have done it too using `ica_use_eog_detection = True`, but we decided against it
4. the reasons are (do like 3 main points. e.g. veog showed no correlation with blinks for some subjects. amplitudes/offsets/excursions were too large, ...)

]

[summarize all this] EOG is excluded from scalp-channel PyPREP, the EEG ICA fit and the scalp average reference. Automatic EOG-based component selection remains disabled

(`ica_use_eog_detection = False`): the detector first chooses an EOG channel using filtered energy, so heterogeneous channel gains can affect that selection even though a correlation coefficient itself is invariant to fixed positive scaling. There is no arbitrary auxiliary amplitude cutoff or invented rescaling. Enabling `ica_use_eog_detection = True` would be the pipeline counterpart to selecting components by their relationship to EOG, followed by manual verification as in the paper. We instead use ICLabel and visual EEG-component review. In subject 014, visual inspection additionally suggested possibly exchanged EOG roles, with blink-like spikes apparent in HEOG. This is an inspection-based suspicion, not a verified wiring diagnosis; the channel labels and samples were left unchanged. EOG was used only as auxiliary diagnostic evidence, not as an automatic cleaning input. The supplied auxiliary values show large offsets and excursions in many recordings, with substantial variation between files. The following summary uses the conventional EEGLAB-to-SI scale applied by the loader. It does not establish the original physiological gain. For each recording, the 99th percentile is calculated from the absolute deviation from that recording's median; this measures excursions without allowing a DC offset to dominate.

Channel	Recording-median offset range (V)	Median 99th-percentile excursion (mV)	99th-percentile excursion range (mV)
VEOG	-1.603 to 2.992	135.3	0.571 to 514.4
HEOG	-1.515 to 1.126	101.3	0.246 to 3,390.3

Table 1: Auxiliary values across 34 recordings per channel under the supplied scaling convention. The excursion columns summarize one 99th-percentile value per recording; they are not pooled-sample percentiles.

These values are not uniformly large: some auxiliary channels have much smaller variation. Recording peak-to-peak values reach 4.22 V for VEOG and 5.48 V for HEOG, while the lowest values are approximately 1.95 and 1.04 mV. Thus neither a single representative trace nor one correction factor describes the entire cohort. A direct sample audit found that MNE's values agree with the native `.fdt` values after the expected factor of 10^{-6} . This does not support a second, accidental scaling error in the current loader. Acquisition headers and a documented auxiliary gain are absent. The manufacturer's guidance describes auxiliary EOG arrangements whose output scaling can differ from scalp EEG; a gain mismatch is plausible, but the available files do not establish the precise hardware configuration or a universal correction. The values remain unchanged. [8]

[summarize this as well for above]

Low-variation VEOG channels

In the audit's selected 120 s intervals, VEOG for sub-003, sub-004 and sub-006 showed both near-zero association with the frontal comparator Fp1–Pz and unusually small filtered variation relative to other VEOG channels. Sub-001 instead showed a strong frontal association. The audit used a 0.5–15 Hz bandpass on those intervals, independently of the main pipeline. This can reflect poor auxiliary signal, an uninformative interval, or another acquisition difference. It does not by itself prove that a channel is unusable throughout both recordings. A matched visualization against sub-001 and the frontal EEG comparator is retained in `TOD0.md`. ICLabel provides EEG component proposals for inspection of component maps, spectra and time courses. Auxiliary traces remain available for normalized, per-channel inspection; they are useful timing evidence where a physiological association is visible, without treating their current amplitudes as calibrated EOG. The full audit, segment definitions and per-recording results are in `docs/eog_investigation.md`.

Second pipeline pass: from ICA models to analysis epochs

After the computation of the ICA models, we have continuous 250 Hz data, 1–100 Hz filtered and epoched to fixed windows. The analysis instead needs 500 Hz, 0.1–40 Hz EEG aligned to specific events, followed by trial cleaning and a prestimulus baseline. We therefore use a separate pipeline configuration, `config_postica.py`, reusing the reviewed models from the previous pipeline execution.

1. Select eligible events from the prepared dataset

The original paper notes that some of the game events occur in quick succession, not giving each of the game events a meaningful epoch. A look at the data confirms this issue: [picture of a segment where many events happen within the window, and the eeg reaction seems to be most tied to the first of those events. also it should be evident that a 500ms window is enough to circumvent this issue]

The issue affects wall crashes (91) and star collection (93). To circumvent this issue each of these events is compared to the last retained event of the same type: if the time between these events is below 500 ms, this repeat event gets marked as excluded. This is done by adding a `EXCLUDED_REPEAT_...` prefix to the these affected wall/star events, allowing e.g. ERPs to ignore these events. The same effective step was done in the original paper as well and affects 3,010 / 4,443 wall events and 6,556 of 13,776 star events.

2. Filter, reference and epoch the EEG

The original 500 Hz runs have to be filtered continuously to 0.1–40 Hz. Then, with `task_is_rest = False`, the new pipeline creates `-0.2...+0.8 s` epochs around the ten eligible event labels to supply the intervals used by our classifiers.

3. Apply the reviewed ICA exclusions

The pipeline loads the saved models and native component-status tables and applies the 193 approved exclusions to compatible continuous EEG and event epochs. EOG samples remain unchanged.

4. Repair or reject analysis epochs

AutoReject is now run again. Previously, it was used before ICA to select training windows. This time, using `reject = "autoreject_local"`, it can repair individual channels in retained epochs as well as reject complete trials. The following counts describe this second stage.

Of 36,258 events that are eligible for the classifiers, 13 windows extend beyond a recording boundary. AutoReject rejects 4,049/36,244 (11.2%) epochs. Among the retained epochs, **28,316**

(88.0%) receive at least one channel repair. The pooled median is four repaired channels per retained epoch; subject medians range from one to nine. Selected interpolation limits are 4, 8 or 16 (default).

[put subject names on the ticks for the top half as well]

Condition	Eligible Events	Final trials
Oddball Standard	2,719	2,247
Oddball Rare	681	539
Gambling Win	618	515
Gambling Loss	617	494
Player Crash Wall	1,433	1,235
Collect Star	7,220	6,457
Missile Hit Enemy	6,495	5,923
Player Crash Enemy	1,810	1,556
Collect Ammo	1,103	1,007
Shoot Button	13,562	12,222

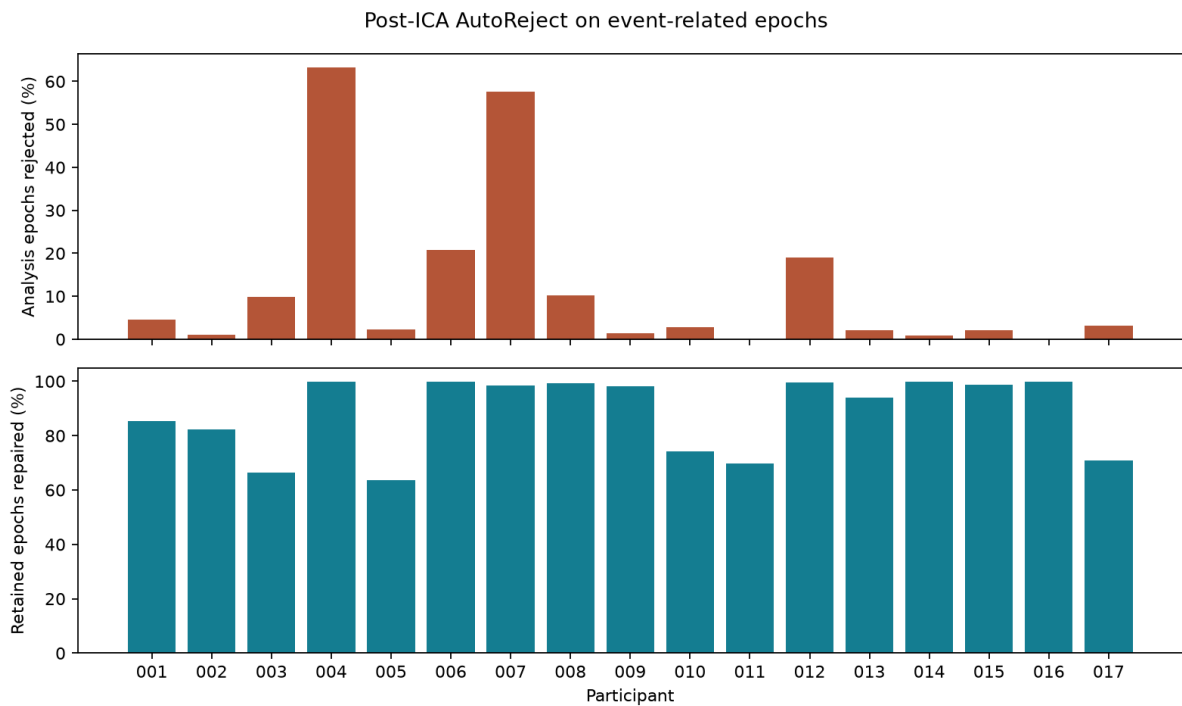


Figure 8: Post-ICA trial rejection and the percentage of retained trials with at least one channel repair.

The rejection is uneven across subjects: 63.2% for 004 and 57.6% for 007. Subject 007 retains only 5/44 rare oddball trials and 34/156 standards; 004 retains 11/40 gambling losses and 17/40 wins. Subject 001 retains only 87/200 oddball trials despite 4.6% rejection pooled across all conditions (this is due to gameplay being a much larger contributor of epochs than oddball or gambling).

We again experimented with lowering rejection to retain more trials, but did not see an improvement in data quality.

5. Baseline correction and saved analysis outputs

After repair/rejection, the pipeline applies the $-0.2...0$ s baseline (same as the original paper) and finishes the final *_proc-clean_epo.fif files reports. Global bad channels remain marked in these epochs.

ERP checks before and after cleaning

The following figure compares the ERPs for exemplary trials before and after our main cleaning. For the Oddball task, it can be seen that the Standard event stays much closer to the baseline after cleaning, and both events no stay much more stable after 0.6 s. For the Gambling task, cleaning mostly corrects the dip seen at ~ 0.5 s from -4 uV to 0 uV.

[remove the baked in text at the bottom of the image]

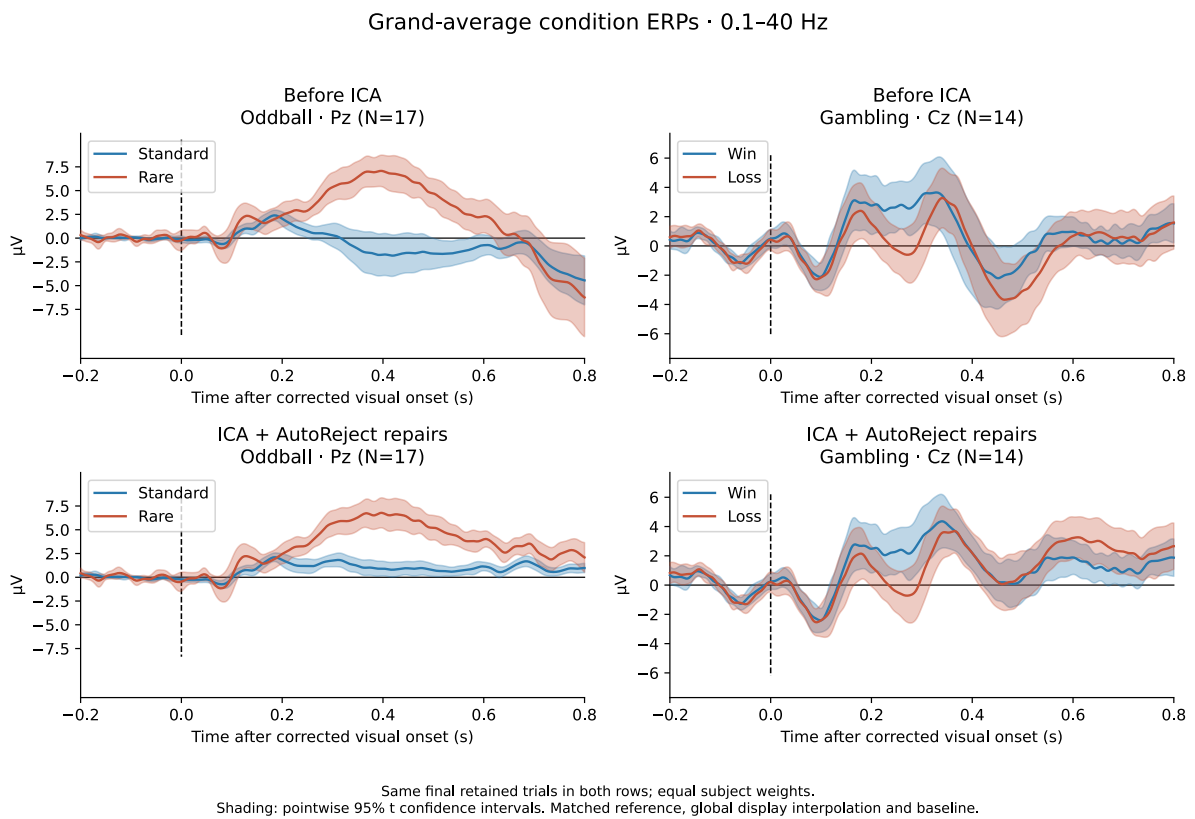


Figure 9: Grand-average condition ERPs before ICA (top) and after ICA plus final AutoReject repairs (bottom), at Pz for oddball (left, N = 17) and Cz for gambling (right, N = 14). Both rows use identical final retained trial IDs, 0.1–40 Hz filtering and $-0.2...0$ s baseline. Shading is the pointwise 95% t confidence interval

Analysis

Replication objective

The original study examined whether EEG patterns associated with exemplary events could be identified during continuous and complex gameplay. To do this, the authors trained participant-specific classifiers on oddball and gambling trials, then transferred them to game events. Missile hits, enemy crashes and wall crashes showed a bias towards rare-oddball classifications at later post-event times (P3). Missile hits also resembled gambling wins. However, the expected loss-like response to crashes could not be established, and crashes were partly also classified as win-like. The authors interpreted these findings as evidence that laboratory signatures associated with salience and reward generalize to gameplay. [1]

Our objective is to see whether these relationships remain observable after our different preprocessing pipeline and with EEGNet. We therefore examine whether gameplay events are assigned the same laboratory categories as in the paper, and whether these classifications differ from those of ordinary gameplay.

Analysis design

We use leave-one-participant-out evaluation. For each participant, a classifier is trained on the other participants and evaluated on the excluded participant's laboratory and gameplay trials. The original paper trained and transferred classifiers within the same participant. Our analysis therefore tests generalization across participants as well as across tasks; it is not a direct replication of the original training design.

We examine the complete 0–800 ms response and two shorter intervals: 300–600 ms for oddball and 200–350 ms for gambling. The shorter intervals focus on the P3 and feedback-related responses relevant to the original findings. Both analyses use the same participant assignments and EEGNet model defaults.

Classifier choice

We use EEGNet, a compact convolutional network designed to learn spatial and temporal EEG features from voltage samples. This allows us to evaluate cross-participant classification without restricting the model to a predefined electrode or ERP amplitude measure. We use the Braindecode 1.5.1 implementation with its default model parameters, supplying only the input dimensions, sampling frequency and number of output classes. Training settings are specified separately in the methods appendix. [9], [10]

Input data and participant inclusion

The cleaned dataset contains 2,786 oddball trials from 17 participants: 2,247 standard and 539 rare. Fourteen participants contribute 1,009 gambling trials: 494 losses and 515 wins. These labelled laboratory trials train the direct classifiers. Gameplay events provide the transfer dataset and do not contribute additional rare/standard or win/loss training labels.

We use the repaired 0.1–40 Hz epochs with the –200 to 0 ms baseline. Globally bad channels are interpolated in analysis copies to obtain 63 common EEG inputs; EOG is excluded. Polyphase resampling reduces the sampling rate from 500 to 250 Hz. The existing reference, local repairs and event corrections are retained.

All participants with both laboratory labels are included. A sensitivity summary restricts evaluation to test participants with at least 20 trials per class, addressing the imprecision of performance estimates from small test sets. This pragmatic threshold changes only the reported subset, not the fitted models or the participants used for training.

Whole-epoch classification and transfer: 0–800 ms

For each held-out participant, three other participants form an inner validation set for selecting the EEGNet training duration. The final model is then initialized afresh and fitted to all participants except the test participant. Scaling is estimated from training data only, and each participant/class combination receives equal total training weight. Detailed settings are provided in the methods appendix.

Laboratory performance is measured by balanced accuracy, the mean recall across the two classes, with 50% as the no-discrimination reference. Gameplay transfer is summarized as the fraction of trials assigned rare or win at a fixed 0.5 decision threshold. These fractions are compared with predictions on randomly sampled gameplay epochs. The controls represent ongoing gameplay and may contain recorded events. Gameplay comparisons require at least 20 event and control trials per participant. Results are calculated within participants and then averaged with equal participant weight.

Laboratory classification

Task	Participants	Balanced accuracy	Descriptive 95% interval
Oddball	17	80.5%	76.3–84.3%
Gambling	14	66.6%	60.8–72.3%

EEGNet discriminates the laboratory conditions across held-out participants, with higher mean accuracy for oddball than gambling. Restricting the summary to participants with at least 20 trials per class gives 81.9% for oddball ($N = 15$) and 65.2% for gambling ($N = 11$). The overall pattern is therefore similar after excluding the smallest test sets from the summary.

Gameplay transfer

The table reports whole-epoch EEGNet predictions. Control means use the same participants as the corresponding event comparison; differences are expressed in percentage points (pp).

Label	Game event	N	Event labels	Gameplay controls	Difference (pp)
Rare	Missile hit	17	28.7%	18.2%	+10.5
Rare	Enemy crash	17	42.1%	18.2%	+23.9
Rare	Wall crash	16	31.5%	17.9%	+13.6
Win	Missile hit	14	65.5%	51.6%	+13.8
Win	Enemy crash	14	70.7%	51.6%	+19.0
Win	Wall crash	13	61.0%	51.8%	+9.1

All three game events receive rare labels more frequently than random gameplay, although their mean rare-label fractions remain below 50%. This supports a relative shift towards the rare category, but does not reproduce the paper’s time-specific predominance of target classifications.

Missile hits also receive more win labels than random gameplay. The same direction is observed for enemy and wall crashes, limiting the interpretation of the gambling classifier as a selective measure of reward. The original paper similarly did not establish the predicted loss-like response to crashes.

The event-minus-control differences are positive for all six comparisons, and their descriptive 95% intervals exclude zero. These intervals summarize variation across participants conditional on the fitted models. Because the models share training participants, they should not be interpreted as independent confirmatory tests.

Using the complete 800 ms response does not localize the information responsible for these shifts. The shorter intervals below assess whether the pattern is also present in the response periods relevant to the original study.

Comparison in the P3 and feedback intervals

We also train EEGNet using only 300–600 ms for oddball and 200–350 ms for gambling. These intervals focus on the P3 and feedback-related responses. The classifiers use the same default model settings, participant assignments and training procedure as the whole-epoch analysis. This comparison is exploratory.

The original paper used short moving windows and participant-specific classifiers. Our broader, fixed intervals and cross-participant training address the general transfer pattern without reproducing the original time courses. We examine both the proportion of rare or win labels and its difference from random gameplay.

Laboratory classification and gameplay transfer

Within these intervals, EEGNet reaches 73.8% balanced accuracy for oddball and 61.2% for gambling. The descriptive 95% intervals are 69.0–78.5% and 58.3–64.3%, respectively.

The table reports transfer from the direct rare/standard and win/loss classifiers. Event and control means use matched participant sets.

Label	Game event	N	Event labels	Gameplay controls	Difference (pp)
Rare	Missile hit	17	29.6%	17.7%	+11.9
Rare	Enemy crash	17	41.3%	17.7%	+23.6
Rare	Wall crash	16	31.7%	17.2%	+14.5
Win	Missile hit	14	65.1%	48.3%	+16.8
Win	Enemy crash	14	74.3%	48.3%	+26.0
Win	Wall crash	13	63.3%	48.3%	+15.1

For oddball, all three events shift towards rare relative to random gameplay, but their mean rare-label fractions remain below 50%. The predominantly target-like result is therefore not recovered by the cross-participant models in the selected 300–600 ms interval. For gambling, hits receive predominantly win labels at 200–350 ms, consistent with the direction of the paper’s direct win-versus-loss result. Both crash conditions also receive predominantly win labels.

Missile hits receive 65.1% win labels, with a descriptive 95% interval of 57.9–72.2%. The difference from controls is +16.8 percentage points, with an interval of +10.1 to +23.2 points. As with the whole-epoch analysis, all six event-minus-control intervals exclude zero. These are descriptive summaries of the fitted models.

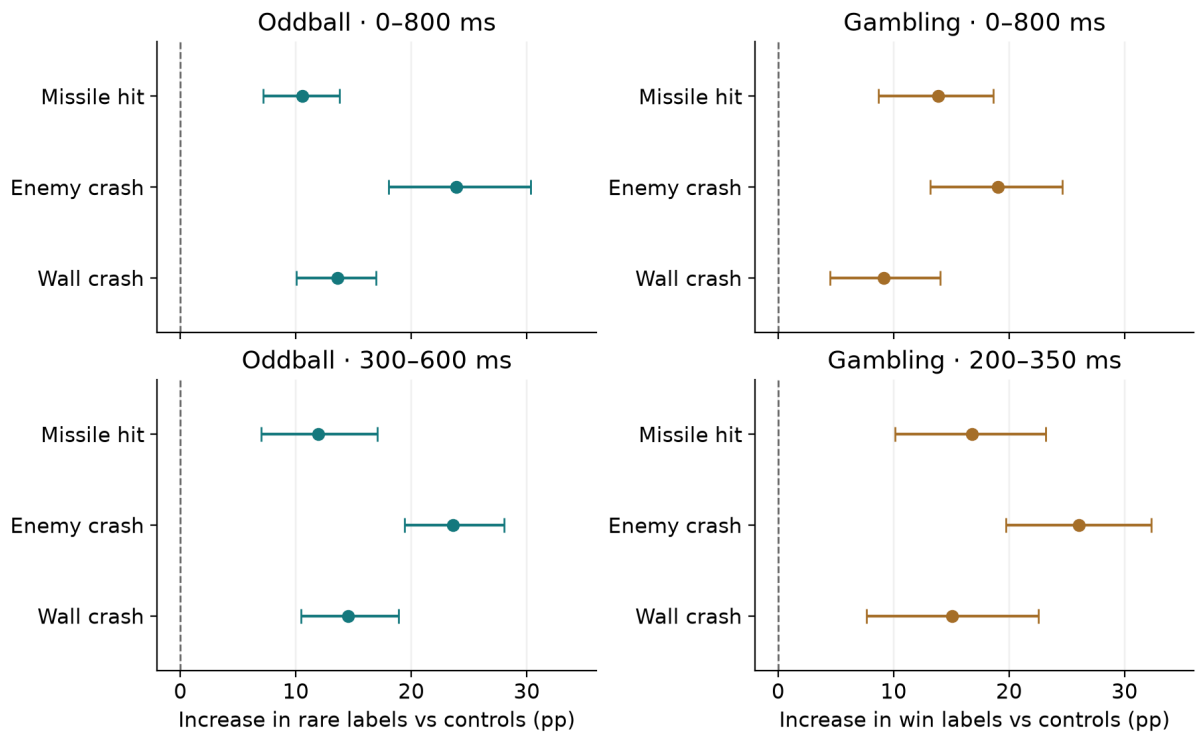


Figure 10: EEGNet transfer using the whole response (top) and the shorter response intervals (bottom). Points show the mean within-participant difference between event and control label fractions; bars are descriptive 95% participant bootstrap intervals. Positive values indicate more rare or win labels than for random gameplay.

Laboratory-versus-gameplay contrasts

The paper also compared each laboratory category with random gameplay. We therefore train separate EEGNet classifiers for rare versus gameplay, standard versus gameplay, win versus gameplay and loss versus gameplay in the same shorter intervals. Training uses laboratory trials and random gameplay epochs from the training participants only; the test participant remains excluded.

The table shows held-out classification performance for these contrasts and the fraction of game-event trials assigned the laboratory label. Each row represents a separate binary classifier, so the label fractions across rows are not complementary probabilities.

Laboratory category	Source balanced accuracy	Hit labels	Enemy-crash labels	Wall-crash labels
Rare	80.7%	26.2%	34.8%	27.1%
Standard	75.6%	31.4%	28.9%	26.0%
Win	74.7%	35.1%	41.7%	31.1%
Loss	74.2%	24.9%	21.5%	22.4%

Although these classifiers discriminate laboratory trials from gameplay, all mean laboratory-label fractions for the game events remain below 50%. Missile hits receive 35.1% win labels in the win-versus-gameplay model, compared with 19.4% for random gameplay. The loss-versus-gameplay model gives hits 24.9% loss labels, compared with 15.6% for controls. Both laboratory-label fractions therefore increase relative to controls, while gameplay remains the majority classification.

Agreement with the original hit result is consequently limited to the direct win-versus-loss contrast in this interval. The additional models do not recover a predominance of laboratory classifications

over gameplay. They may also exploit differences between the laboratory and gameplay contexts rather than a specific event response.

Reproducible deliverables

This report presents the selected preprocessing, EEGNet classification with default model parameters, and gameplay transfer using whole responses and shorter response intervals. The source is `report/report.typ`; compile it from the project root with `typst compile --root . report/report.typ report/report.pdf`.

`README.md` documents execution and review commands. The configuration, package/version records, preparation manifest, per-recording PyPREP files and `artifacts/qc/` tables preserve the inputs and intermediate evidence. The companion `docs/paper_and_decisions.md` contains the narrower technical decision record. `docs/course_requirements.md` summarizes the assignment, and `docs/lecture_overview.md` maps the lecture material to the processing choices.

The published article defines the scientific question and original method. [1] The shared dataset and local event tables define the observations available for this reanalysis. [2] The report follows the article where the provisional scripts disagree and documents implementation-level ambiguities in the appendix.

Discussion

Comparison with the original findings

EEGNet discriminates the laboratory conditions across held-out participants and shows event-related shifts in gameplay classifications. The comparison in the P3 and feedback intervals gives different outcomes for oddball and gambling.

For oddball, gameplay events shift towards rare labels relative to controls, but the mean fractions remain below 50% at 300–600 ms. We therefore do not recover the original predominance of target classifications in this analysis. This result concerns the selected interval and cross-participant models; it does not establish an absence of attention-related responses during gameplay.

For gambling, hits receive predominantly win labels at 200–350 ms and more win labels than random gameplay. This agrees with the direction of the original direct win-versus-loss result. However, the separate win-versus-gameplay classifier assigns most hit trials to gameplay, so agreement does not extend to the full set of original contrasts. Crashes also shift towards win, limiting a selective reward interpretation. The original paper similarly failed to establish its predicted loss-like crash response and reported some win-like classifications for crashes. [1]

These findings constitute partial agreement in a reanalysis of the same dataset. The differences in preprocessing, classifier type, training population and temporal intervals prevent attribution of the discrepancies to any one methodological choice.

Limitations

Transferred labels indicate similarity according to the learned classifier, not direct evidence of attention, reward or punishment. EEGNet may use shared amplitude features, sensory or motor activity, differences between tasks, or residual artifacts. The analysis does not establish which features drive its decisions or whether they identify the same psychological process in both settings.

The participant sample is limited to 17 for oddball and 14 for gambling. Large gameplay trial counts improve within-participant estimates but do not increase the number of independent participants or labelled laboratory training examples. Several participants have few retained trials in one laboratory category, and rejection and interpolation vary across recordings.

Participant exclusion applies to classifier fitting and selection. Preprocessing was completed retrospectively using each participant's whole recording, so the results are conditional on that cleaning procedure. They do not represent a fully independent evaluation from unseen raw data to prediction. Furthermore, the held-out models share training participants. The reported bootstrap intervals do not account for all dependence between model results or variability from retraining; each network uses one fixed initialization.

The analysis is exploratory and uses the original dataset rather than an independent replication sample. The fixed intervals do not provide a complete time course, and neighbouring gameplay events may contribute to the measured responses. Positive transfer shifts therefore support further investigation rather than a definitive identification of canonical neural events.

Further work

A matched within-participant EEGNet analysis would bring the training design closer to the paper and help assess the effect of pooling participants. The paper's shorter moving windows and derivative representation could then be examined as additional methodological comparisons. Stronger claims about psychological specificity would require controls that distinguish sensory and motor responses from salience or reward, followed by evaluation in an independent sample.

Conclusion

Using EEGNet with default model parameters, we identify laboratory EEG distinctions that generalize across participants and observe shifts in gameplay classifications. The feedback-interval analysis agrees with the direction of the paper's win-versus-loss result for missile hits. It does not recover predominantly target-like classifications in the selected oddball interval or predominantly laboratory-like classifications against gameplay controls.

The results therefore show partial agreement with the original transfer findings. Win-like crash responses and uncertainty about the features used by EEGNet limit their interpretation as specific measures of attention, reward or punishment.

Appendix: classifier settings and checks

Inputs and gameplay controls

Analysis copies of the native repaired epochs retain the 0.1–40 Hz band, –200 to 0 ms baseline, reference and local trial repairs. Globally bad channels are spherically interpolated to give 63 common scalp inputs; EOG is omitted. Polyphase resampling gives 250 Hz, with 200 samples from 0 to 796 ms for the whole-epoch analysis. Neither the +40 ms event correction nor the 500 ms repeat-event rule is applied again.

Up to 400 control anchors per participant are sampled on randomly phased 2 s grids within event-supported gameplay spans; gaps above 30 s split spans. Controls may overlap recorded events and represent ongoing gameplay. AutoReject estimators recovered from the ICA-only short epochs reproduce the original rejection masks and checked repaired voltages. The saved estimators repair or reject controls before the same baseline, interpolation and resampling. This retains 6,049 of 6,800 candidates. Estimators, repair logs and trial IDs are retained.

Every source model includes all eligible participants with both training labels. The separate trial-count sensitivity summary uses at least 20 trials per class in the test participant; it does not exclude training participants or trigger model refitting. Gameplay contrasts require at least 20 retained event and control trials per participant. Subject 007 has only ten wall crashes, giving 16 oddball and 13 gambling participants for wall-crash comparisons.

EEGNet architecture and training

We instantiate Braindecode 1.5.1 EEGNet as `EEGNet(n_chans=63, n_outputs=2, n_times=n_times, sfreq=250)`. All optional model arguments use their defaults: $F1 = 8$ temporal filters, depth multiplier $D = 2$, $F2 = 16$ separable filters, a 64-sample temporal kernel and dropout 0.25. The whole-epoch network has 2,306 trainable parameters. These are implementation defaults, not parameters shown to be optimal for this dataset. [9], [10]

Training uses Adam with learning rate 0.001, weight decay 0.0001 and batch size 64. These are explicit training choices; they are not specified by the EEGNet model constructor. Trial weights give each participant/class combination equal total loss weight. Channel means and standard deviations come only from the current training data. A model tested on a participant receives none of that person's trials for classifier fitting, scaling, model selection or batch-normalization updates.

For each outer test participant, three other participants provide inner validation and the remaining participants train the candidate network. Weighted validation log loss, with equal total participant/class weight, selects the training epoch count. The search runs for at least 12 and at most 80 epochs, with stopping patience of 12 epochs without improvement. A freshly initialized network is then fitted to all other participants for the selected number of epochs. The selected count can be below 12 because the minimum constrains the search duration, not the epoch with the lowest validation loss.

For the shorter intervals, inputs contain 75 samples from 300–600 ms for oddball and 38 samples from 200–350 ms for gambling. Endpoints are half-open on the 250 Hz grid. Default architecture settings are retained, giving 2,178 and 2,146 parameters. The 64-sample kernel exceeds the gambling input duration; the implementation handles this with zero padding, without receiving EEG from outside the cropped interval. Earlier filtering and baseline correction can still affect temporal interpretation.

The shorter-interval analysis includes three contrasts per task: the direct laboratory distinction, its positive category versus ordinary gameplay, and its negative category versus ordinary gameplay. Positive means rare or win; negative means standard or loss. Laboratory-versus-gameplay classifiers

use random controls only from their training participants. The held-out random controls also appear in their source evaluation and transfer summaries, so these are not independent pieces of evidence.

Summaries and reproducibility

Laboratory summaries average participant balanced accuracies. Gameplay labels use a fixed 0.5 threshold. Each person's event fraction is paired with their ordinary-gameplay fraction before averaging; control means use the same participants as the event comparison. Intervals use 10,000 participant bootstrap resamples and describe uncertainty conditional on fitted models. They are not simultaneous intervals, do not refit networks and do not account for all dependence arising from shared training participants. No formal confirmatory significance claim is made.

The analysis comprises 31 whole-epoch and 93 shorter-interval EEGNet fits. Saved-model checks independently reproduce every laboratory and gameplay prediction and verify trial identities, participant separation, default settings, training-only scaling and validation-based epoch selection. These checks establish implementation consistency, not psychological specificity.

The classifier code is in `analysis/eegnet_defaults/`, with the configuration, fitted models, per-trial predictions and audit results in `artifacts/eegnet_defaults/`. Execution and cached-input requirements are documented in the accompanying README; the entry point is `./run-eegnet-defaults`. Report tables and the transfer figure are generated from the audited default-model summaries by `scripts/make_report_eegnet_defaults.py`.

Appendix: earlier preprocessing experiments

The detailed pilot history below is retained to preserve the earlier report's numerical comparisons and methodological evidence. These are historical and sensitivity results, not additional cleaning applied to the selected final cohort.

Earlier ICA workflows and controlled comparisons

The first cohort workflow fitted one extended Infomax model per subject across both runs, with seed 2026, a maximum of 500 iterations and a rank-aware component count. Subject 007 completed the initial execution check before the remaining cohort runs. All 17 historical fits and standard MNE reports were completed. This branch was superseded by the selected native cohort described below.

An exploratory local-autoreject run rejected 23.0% and 33.7% of fitting windows for subjects 001 and 007, with disproportionate loss of oddball/gambling windows. These are diagnostics of the initial fitting-window policy, not final ERP trial-loss rates. Overlapping event-centred windows also submitted the same gameplay samples repeatedly. The first cohort workflow therefore used non-overlapping four-second windows without baseline subtraction, and a fixed gross-artifact screen: reject a window when any remaining EEG channel exceeds 500 μV peak-to-peak after average reference. This threshold is a practical screen, not a demonstrated optimum; retained-window and component quality still require inspection.

In that earlier cohort, original event labels defined oddball, gambling and gameplay blocks wherever they occurred, including subject 008's gameplay tail in run 01. A task-family change or gap longer than 30 s starts a new block. The gap cutoff is an operational inference of inactivity, not a validated break annotation. Blocks receive 2 s of padding, clipped to recording limits and to neighbouring block midpoints when needed. Complete four-second windows are retained and short tails are omitted. The 500 ms repeat-event rule and +40 ms correction belong to event-related analyses and are not applied to these fitting windows. The cohort audit identifies 10,308 candidate windows before amplitude rejection; retained counts are recorded separately by run and task.

Across the cohort, **10,115 of 10,308 fitting windows were retained (98.13%)**, providing 32.9–54.1 minutes of unique training EEG per subject. Rejection was 1.99% for oddball, 1.28% for gambling and 1.89% for gameplay. Subject 003 had the largest overall loss (6.08%); within a task, subject 013 lost 11 of 98 oddball windows (11.22%). These are fitting-window counts, not final ERP trial counts. The lower rejection than the exploratory workflow reflects several changed choices, including window selection and threshold; it is not an isolated comparison of rejection algorithms.

There are **1,003 fitted components**, 56–62 per subject, matching the usable EEG count minus one for average reference. All input/output integrity checks passed, weights are finite and full rank, and every fit met an optimizer stopping criterion before the 500-pass limit. The batch completed in 1 h 26 min, reusing subject 007's preflight fit. Individual numerical fits ranged from 1 min 45 s to 50 min 51 s, with a median of 7 min 46 s. Runtime alone does not indicate component quality.

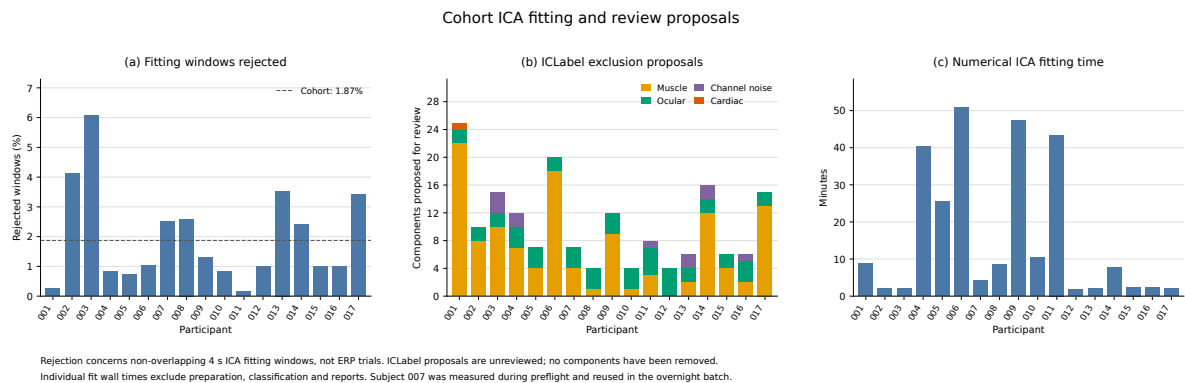


Figure 11: ICA fitting-window rejection, ICLabel exclusion proposals and numerical fitting time by subject. No component proposals were applied in this historical branch. Subject 007's timing comes from the preflight fit reused by the batch.

ICLabel proposed **177 exclusions**: 120 muscle, 45 ocular, 11 channel noise and one cardiac component. The largest proposal counts were in subjects 001 (25), 006 (20) and 014 (16). These counts do not measure the fraction of EEG variance attributable to artifacts. A preliminary screen of 25 component panels from seven subjects found recognizable ocular patterns, plausible brain components and focal fast artifacts. Some high-confidence labels remained ambiguous, including a cardiac proposal without established periodicity in its displayed panel. This supports further component review rather than automatic acceptance. No components were removed in that historical branch.

A closer comparison for subject 001 found that the exploratory and first-cohort report overlays average differently aligned data, so their waveform shapes cannot assess preservation of task responses. Matching component maps and evaluating both models on the same EEG found close counterparts for four initially clear brain components, but weaker narrow alpha peaks in two first-cohort counterparts. The fitting-window and rejection choices therefore remain under sensitivity review; successful numerical execution alone does not establish equal physiological separation.

A controlled subject-001 sensitivity fit kept the current 250 Hz data, window grid, channels, reference and Infomax settings, but selected windows using the pilot's saved channel-amplitude thresholds and consensus rule. It retained 608 of 812 windows. Correlations of the four matched scalp maps with the pilot increased from 0.85–0.97 to 0.98–0.99; comparison on the same EEG also recovered some of the earlier alpha structure. This implicates fitting-data selection, including its effect on task weighting, while showing that 250 Hz and non-overlapping windows can retain close counterparts of those sources. It is a single-subject sensitivity check, not a validated cohort rejection policy; its historical thresholds are not transferred to other subjects.

A second controlled pilot fitted fresh local autoreject to the complete current window sets for subjects 001 and 007, replacing the 500 μV screen while holding the remaining settings constant. It retained 693/814 and 484/599 windows; ICA fitting took 17 min 54 s and 4 min 34 s. Subject 001 more closely recovered the four earlier source patterns (map correlations 0.93–0.99), including much of the original ICA009 alpha structure. Subject 007 retained similar recognizable brain and eye sources, with mixed spectral and artifact-separation changes. Both lost proportionally more oddball than gameplay windows. AutoReject also admitted one 612 μV localized TP10 transient that the fixed screen rejected, illustrating that its channel-consensus criterion is not a strictly stronger amplitude cutoff. Its selected interpolation parameters (16 and 8) did not cause channel interpolation before fitting. These results do not establish a uniform cohort-wide improvement; the earlier completed cohort remained unchanged by that comparison. The controlled comparisons and their limitations are in `docs/ica_pilot2.md`.

That historical batch saved decompositions, fitting-data audits and standard MNE component reports. Its ICLabel probabilities remained proposals for manual review of maps, spectra and time courses; no components were removed automatically in that branch. Fitting-window rejection, component exclusion and final analysis-trial rejection remain separate decisions. Automatic EOG-based exclusion remains disabled while auxiliary calibration is unresolved. All 1,003 native component property panels are present. These display up to 160 windows balanced across runs/tasks; fitting and classification use all retained windows. Their fixed-window averages are not task ERPs. The cohort audit and specific review priorities are in `artifacts/ica_full/review_summary.md`; execution details are in `docs/ica_overnight.md`.

Simplified native fixed-window pilot

A further subject-001 pilot used `task_is_rest=True` to select native non-overlapping four-second windows across the recordings. This is an epoching setting, not a claim of resting-state acquisition. It removes the inferred task boundaries, 30 s gap rule and padding from the processing path. The prepared bad-channel metadata determines the shared channel/reference set; prepared event timing and eligibility are reserved for later ERPs and do not affect this ICA grid. Filtering, 250 Hz resampling, fresh local autoreject, fitting, ICLabel and report generation all use ordinary pipeline stages. The command stops before applying ICA component removal.

This pilot retained 817/904 windows (90.4%) and fitted 60 components in 8 min 49 s; the complete native command took 12 min 49 s. Its numerical ICA settings were unchanged. The largest retained good-channel peak-to-peak amplitude was 311 μV . ICLabel proposed 28 exclusions, written to the standard component-status TSV and saved exclusion list. These require review before the application stage; they are not final manual decisions. All 60 native component panels are present.

Most maps remained similar to the second pilot, but two originally highlighted brain sources had weaker map and alpha correspondence. Their correlations with the original ICA008 and ICA009 maps were 0.82 and 0.85, compared with 0.93 and 0.99 in the second pilot. Spectra were compared on identical EEG windows, so these differences cannot be attributed simply to different report averages. The simpler workflow executes correctly; this comparison does not establish superior source separation or isolate between-task data as the cause, because window alignment and learned rejection also changed. Detailed results and validation are in `docs/ica_native_pilot.md`. The selected native cohort is reported in the Preprocessing chapter.

Controlled ICA solver comparison

A final subject-001 test changed only the solver to Picard configured for extended Infomax, using a separate derivative directory. All 817 retained epochs and their saved voltage samples were exactly identical to the native Infomax run. Picard stopped after 210 iterations without a convergence

warning, but its fit took 16 min 47 s compared with 8 min 49 s for Infomax; complete command times were 20 min 44 s and 12 min 49 s. Both produced 60 components. With one-to-one matching, 57/60 scalp maps correlated above 0.998 and the four highlighted brain-source spectra nearly overlapped. Three components differed more, but all 28 automatic exclusion decisions agreed after matching. This single-subject comparison provides no speed advantage or clear improvement in those sources from switching solvers. The timings use the same thread limits and each solver's native stopping rule; they are not a repeated benchmark. Full comparisons are in docs/`ica_picard_pilot.md`.

References

- [1] J. F. Cavanagh and J. Castellanos, "Identification of canonical neural events during continuous gameplay of an 8-bit style video game," *NeuroImage*, vol. 133, pp. 1–13, 2016, doi: 10.1016/j.neuroimage.2016.02.075.
- [2] J. F. Cavanagh and J. Castellanos, "EEG: Continuous gameplay of an 8-bit style video game." [Online]. Available: <https://openneuro.org/datasets/ds003517>
- [3] Okimoka, "EEG_VL: accompanying project repository." [Online]. Available: https://github.com/Okimoka/EEG_VL
- [4] BIDS contributors, "Brain Imaging Data Structure: Electroencephalography." [Online]. Available: <https://bids-specification.readthedocs.io/en/stable/modality-specific-files/electroencephalography.html>
- [5] PyPREP contributors, "PyPREP: NoisyChannels reference." [Online]. Available: <https://pyprep.readthedocs.io/en/latest/generated/pyprep.NoisyChannels.html>
- [6] B. Ehinger, "Signal processing and analysis of human brain potentials: Filtering (University of Stuttgart, lecture 3, slide 33)."
- [7] MNE-ICALabel contributors, "Repairing artifacts with ICA automatically using ICLabel." [Online]. Available: https://mne.tools/mne-icalabel/stable/generated/examples/00_icalabel.html
- [8] Brain Products, "EOG recording with the actiCHamp system." [Online]. Available: https://pressrelease.brainproducts.com/eog_actichamp/
- [9] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, "EEGNet: A Compact Convolutional Neural Network for EEG-based Brain–Computer Interfaces." [Online]. Available: <https://arxiv.org/abs/1611.08024>
- [10] Braindecode contributors, "Braindecode: EEGNet implementation." [Online]. Available: <https://braindecode.org/stable/generated/braindecode.models.EEGNet.html>